

# Hongshan Guo

Assistant Professor · Department of Architecture · The University of Hong Kong

hongshan@hku.hk | ORCID: 0000-0002-8716-0343 | +852 9677 8596

## Profile

- **Research:** Design science for human–building–climate systems—probabilistic modeling, physics-informed ML, co-simulation, and climate stress-testing for buildings, cities, and public institutions.
- **Flagship impact:** Published correction of century-old 120 W metabolic default in comfort standards (*Energy and Buildings*, 2025), prompting and winning ASHRAE 1959-TRP (PI, awarded Feb 2026).
- **Methods and media:** Digital twins, probabilistic simulation, scenario-conditioned weather generation, and computer vision; translated into teaching, open benchmarks, and cross-disciplinary design pedagogy.

## Selected Design Science Contributions (2025–2026)

- **Standards intervention:** Demonstrated demographic bias in prevailing metabolic defaults used in global comfort standards; resulting ASHRAE 1959-TRP award positions this work for direct standards revision.
- **Playable performance models:** Built co-simulation workflows integrating data-driven comfort models with EnergyPlus, validated across 13 climates for robust stress-testing under climate uncertainty.
- **Design-facing AI infrastructure:** Led AI-enabled pedagogy platforms (Socratic Oracle; AI Design Coach) deployed across Architecture, Geography, and Computer Science to support evidence-based design reasoning.

## Appointments

- Assistant Professor, Department of Architecture, The University of Hong Kong (Sep 2023–present)
- Principal Data Scientist, Bank of New York Mellon (2020–2023)—led end-to-end ML deployment and model governance for daily \$29–43 billion Federal Reserve account-balance forecasting
- Postdoctoral Researcher, Princeton University (2019–2020)

## Education

- Ph.D., Architecture (Architectural Technology & Computational Design), Princeton University, 2019
- M.Eng., Mechanical Engineering, Columbia University, 2013
- B.Eng., Architectural (HVAC) Engineering, Harbin Institute of Technology, 2012

## Research Expertise

**Design Science & Performance Simulation:** Digital twins and co-simulation frameworks, physics-informed machine learning, probabilistic design exploration under uncertainty, building energy modeling (EnergyPlus, Modelica), reduced-order and surrogate models, post-occupancy evaluation, learning-enabled control.

**AI & Computational Methods:** Physics-informed neural networks (PINNs), uncertainty quantification (epistemic/aleatoric), Bayesian inference and calibration, ordinal regression, model predictive control, reinforcement learning, computer vision (thermal imagery segmentation), synthetic data generation, domain adaptation.

**Environmental Systems & Building Science:** Thermodynamics, heat and mass transfer, HVAC systems and control, thermal comfort prediction, sensor-driven analytics and IoT deployment, wearable sensing, building diagnostics.

**Climate Science & Resilience:** Synthetic weather generation (regime-switching models with CORDEX/CMIP), scenario-conditioned climate projections through 2100, infrastructure stress-testing, urban microclimate sensing, thermal equity and heat-wave vulnerability.

**Tools & Implementation:** Python/Julia/Matlab; EnergyPlus, Modelica, Grasshopper/Ladybug; PyMC/Stan for probabilistic programming; Git/Docker/cloud ML pipelines.

## Mentorship & Supervision

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### Doctoral students

- Kanxuan He — PhD (HKU, 2025–) on probabilistic control; 1st Place, ICML 2025 CO-BUILD Smart Building Competition; leading manuscript in revision for *Energy and AI*.
- Yu Chang — PhD (HKU, 2024–) on cross-cultural thermal comfort; first-author review in *Renewable and Sustainable Energy Reviews* (2025) now informing ASHRAE/EN benchmarks.

### MARch thesis advisees (completing 2026)

- Multi-agentic, physics-informed occupant modeling for environment evaluation in architectural design.
- Neuroarchitecture thesis using EEG to study occupant responses to spatial design conditions.

## Funding & Honors

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- ASHRAE Research Project 1959-TRP — PI, “Demographic-Aware Metabolic Rate Defaults for Thermal Comfort Standards,” \$55,000 (15 months; awarded Feb 2026, commencing Apr 2026). Sponsored by TC 4.1 Load Calculations Data and Procedures.
- Competitive internal startup package (HKU; substantial research funding, 2023)
- Teaching Development & Language Enhancement Grant (TDLEG), HKU — PI, “Socratic Oracle: Multimodal Voice-AI Tutor for Evidence-Based Learning,” HK\$1,000,000 (2025–2028; 31 months)
- Teaching Development Grant (TDG), HKU — PI, “AI Design Coach: A Conversational Design Narrative Builder,” HK\$300,000 (2026–2028). Cross-Faculty deployment across Architecture, Geography, and Computer Science.
- ICML 2025 CO-BUILD Smart Building Competition: 1st Place (faculty mentor/author)
- NeurIPS 2025 Urban AI Workshop (Buildings Challenge): 3rd Place (faculty mentor)
- Gartner “Eye on Innovation” Award (Advanced Solutions, BNY Mellon), 2021
- Lowry Methodology Award, International Conference of Urban Climate (AMS), 2018

## Patents

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- Methods and Systems for Generating Predictions Based on Time Series Data Using an Ensemble Model. US 2022/0300860 A1 (published 2022). Co-inventor.
- Wearable Metabolics: Human Heat Dissipation Sensor via Non-Intrusive Core and Skin Temperature Measurement with Custom-Fitting Wearables. US Provisional 63/300,130 (filed 2022). Co-inventor.

## Selected Publications

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### Journal articles (selected, published)

1. Guo H., Anselmo S., Ferrara M., Niu S., Chi B., Wang X. *Automated Urban Energy Assessment: From Thermal Flyover to AI-Driven Retrofit Prioritization for Sustainable Cities*. *Sustainable Cities and Society*, 148:107589, 2026. doi:10.1016/j.scs.2026.107589. [Thermal flyover, EPC-derived EUI, anomaly triage]
2. Guo H., Aviv D. *From Seven Points to Probabilities: Ordinal Learning for Risk-Aware Thermal Comfort Prediction*. *Building and Environment*, 295:114426, 2026. doi:10.1016/j.buildenv.2026.114426. [Ordinal learning, risk-aware prediction]
3. Guo H., Pigliautile I., Chang Y., Qiao Q., Li Y. *Toward Smarter HVAC Control: Machine Learning Reveals Hidden Drivers in Thermal Comfort Databases*. *Energy and AI*, 24:100709, 2026. doi:10.1016/j.egyai.2026.100709. [Sensitivity analysis, MRT, data quality]

4. Cen C., Guo H., Chew L.W., Wong N.H. *Experimental Study on Gender Differences in Thermal Comfort and Physiological Responses in Fan-Assisted Cooling Environments*. *Energy and Buildings*, 360:117365, 2026. doi:10.1016/j.enbuild.2026.117365. [Fan-assisted cooling, gender differences]
5. Guo H., He K. *Input Quality, Not Statistical Complexity, Determines Climate-Adapted Weather File Fidelity: A Causal Decomposition of Degree-Day Errors*. *Energy*, 353:140867, 2026. doi:10.1016/j.energy.2026.140867. [Weather file validation, causal analysis]
6. Guo H., He K. *Scenario-Conditioned Actual Meteorological Years (sAMY): A Stochastic Weather Generator Using Multi-Decadal Observations*. *Energy and Buildings*, 364:117508, 2026. doi:10.1016/j.enbuild.2026.117508. [Synthetic weather, scenario-conditioned]
7. Guo H., Chang Y., Li Y., Zhou Y., Qiao Q., Lai C.Y., Schuldenfrei E. *Ventilation–Energy Trade-offs in Retrofitted Hong Kong Wet Markets*. *Energy and Buildings*, 355:116918, 2026. doi:10.1016/j.enbuild.2025.116918. [Field measurement, retrofit, IEQ–energy trade-offs]
8. Guo H., Sun R., Xu Y. *Correcting the 120-Watt Assumption: Demographic-Aware Metabolic Rates for Energy Savings and Thermal Comfort Equity in Buildings*. *Energy and Buildings*, 2025. doi:10.1016/j.enbuild.2025.116525. [Standards impact, equity; prompted ASHRAE 1959-TRP]
9. Guo H., He K., Xu Y., Lei Y. *A Co-Simulation Methodology for Integrating Data-Driven Thermal Sensation Models with Building Energy Control*. *Energy and Buildings*, 353:116745, 2026. doi:10.1016/j.enbuild.2025.116745. [Digital twins, co-simulation, 13 climates]
10. Guo H., He K., Luo Y., Chang Y. *Physics-Informed Neural Networks for Robust Thermal Comfort Prediction: Overcoming Data Quality Limitations Through Physiological Constraints*. *Building and Environment*, 2025. doi:10.1016/j.buildenv.2025.113588. [PINNs, physiological constraints]
11. Chang Y., Guo H., Li Y., Chi B., Pigliautile I. *A Data-Driven Qualitative Review of Thermal Comfort Studies: Bridging the Gap Between Western and Eastern Perspectives*. *Renewable and Sustainable Energy Reviews*, 2025. doi:10.1016/j.rser.2025.116020. [Data harmonization, benchmarks]
12. Guo H., Ferrara M., Coleman J., Loyola M., Meggers F. *Simulation and Measurement of Air and Mean Radiant Temperatures in a Radiantly Heated Indoor Space*. *Energy*, 193:116369, 2020.
13. Guo H., Aviv D., Loyola M., Teitelbaum E., Houchois N., Meggers F. *On the Understanding of Mean Radiant Temperature Within Indoor and Outdoor Environments: A Critical Review*. *Renewable and Sustainable Energy Reviews*, 2019.

#### Book chapters

- Guo H., Li Y., Zhou Y., Chang Y., Lai C.Y. *From Open Air to Air-Tight: Analyzing the Ventilation Overhaul in Hong Kong’s Wet Markets and Its Implications*. In *Multiphysics and Multiscale Building Physics*, Springer, 2025.
- Guo H., Coleman J., Gullapalli I. *Accelerating NZEB Design Optimization Through LLM-Based Standardization and Compliance Checking*. In *Multiphysics and Multiscale Building Physics*, Springer, 2024.

#### Refereed conference and workshop papers (selected)

- He K., Guo H. *Data-Driven Cross-Regional Climate Zone Classification for Building Codes: A Future-Compatible Framework*. *E3S Web of Conferences*, 716:10011, 2026. doi:10.1051/e3sconf/202671610011. [Climate zoning, building codes]
- He K., Guo H. *A Temporal Features-Enhanced Mixture-of-Experts Approach for Indoor Temperature Prediction*. *ICML 2025 CO-BUILD Workshop (Oral); 1st Place, Smart Building Competition*. [MoE, temporal modeling, control]

- Niu S., Guo H., Ferrara M., Anselmo S. *Thermal-SAM: Adversarial Prompt-Based Unsupervised Building Segmentation in Thermal Aerial Imagery*. ICML 2025 CO-BUILD Workshop (Poster); **3rd Place, NeurIPS 2025 Urban AI Workshop (Buildings Challenge)**. [Computer vision, envelope diagnostics]
- Guo H., Chang Y., Hu D. *ML-Driven Sensitivity Analysis for Lean HVAC: New Insights from Large-Scale Comfort Data*. ICML 2025 CO-BUILD Workshop (Poster).
- Guo H., Zhou Y., Lai C.Y., Ren C. *From Open Air to Air-Tight: Ventilation Overhaul in Hong Kong Wet Markets*. IBPC 2024 (International Building Physics Conference), Toronto.
- Guo H., Coleman J., Gullapalli I. *Accelerating NZEB Design Optimization Through LLM-Based Standardization and Compliance Checking*. IBPC 2024.

### Manuscripts under review (selected)

#### *Climate & resilience:*

- Guo H., He K. *Controller-Agnostic Benchmarking Across Five Climates: Comfort, Energy, and Peak Trade-offs to 2100*. Under review, Engineering Applications of Artificial Intelligence (special issue), 2025.

#### *Control & learning:*

- Guo H., He K., Xu Y., Shi Z., Aviv D. *Probabilistic Thermal Comfort for Energy-Efficient Building Control: A Risk-Aware Framework*. Under review, Energy Conversion and Management, 2026.

## Teaching & Mentorship

### Courses designed and taught (HKU):

- ARCH7476: Evidence-Based Generative Design — Data-driven decision making in architecture; students frame design problems as testable claims, collect evidence, and produce reproducible analyses with visual communication artifacts. Dual pathways: low-code tools for accessibility, Python/Colab for depth. Mixed MArch/UG/PhD (2025–).
- DESN2003: Research for Innovation — Research methodology from question framing through evidence gathering to communication; parallel tracks for frontier research and applied/buildable outputs. UG (2023–).
- CCGL9065: Our Response to Climate Change | Hong Kong 2100 — Climate action through critique, debate, and speculative design; culminates in a public interactive art & science exhibition envisioning Hong Kong's 2100 landscape. UG (2024–).

**Prior teaching:** Princeton mini-courses on IoT sensing (deploying custom thermal/humidity sensors in campus buildings) and thermodynamics labs (hands-on measurement of radiant environments), 2015–2018.

**Pedagogy:** Evidence-based design workflows combining low-code tools with Python/Colab notebooks; emphasis on reproducibility, uncertainty quantification, and visual communication; typical outputs include scenario videos, interactive dashboards, and spatial equity narratives.

## Professional Service & Collaboration

**Standards engagement:** ASHRAE Research Project 1959-TRP (metabolic rate standards revision)—PI, awarded Feb 2026; published research directly prompted the RFP.

**Reviewing:** CVPR, ICML, NeurIPS workshops; journals including *Energy and Buildings*, *Building and Environment*, *Applied Energy*, *Energy*, *Journal of Building Engineering*, *Sustainable Cities and Society*, *Urban Climate*.

**Leadership:** Co-Chair, Technical Development & Proceedings Integration, CAAD Futures 2025.

**Industry engagement:** Hong Kong Green Building Council (HKGBC), Electrical and Mechanical Services Department (EMSD); prior industry role leading ML productization at Bank of New York Mellon (2020–2023).

**Interdisciplinary collaborations:** Climate scientists, control theorists, computer vision researchers, urban planners, public health scholars; architecture schools including UPenn, Princeton, Politecnico di Torino, University of Perugia, National University of Singapore, University of Tokyo.

## **Creative Practice & Public Scholarship**

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- Exhibitor (co-author), *Social Condenser Extraordinaire: Hong Kong's Municipal Services Buildings*, T1-02, *Projecting Future Heritage: A Hong Kong Archive* (Collateral Event of the 19th International Architecture Exhibition — La Biennale di Venezia), Campo della Tana, Venice, May–Nov 2025.
- Lead instructor, *Our Response to Climate Change | Hong Kong 2100*: public student exhibition combining speculative design with climate narratives and interactive media (HKU, 2024–).